

خلاصة تكيينكـل Integrated و Professional

كـلمـتـين مـهـمـينـ:ـ

- اللاتين نفس المتطلبات عشان كده حطـلهم نفس المطلوب وحتـ لينكات المرشحة من iti وحاجات مني كترشيع حنضيف هنا علي الدوك اي تحديث.
- حاولت اساعد نفسي واساعدكم انه نجمع الدنيا بعيد عن اللغـبطه والـداتا السايحه فياريت دعوـه منكم ليا انه اتقبل .
- لو عاوز تضيف او تعدل حاجه ابعت طلب وحديك صلاحيه .

ملخص الموجود هنا :

- شرح للمتطلبات واساله عليها.
- تجميع للاسـاله الي جات للناس سواء نفس تراكنا او تراك ثاني بس واخذين حاجه مشتركه معانا زي لغة برمجه او دتا ييز مثلا.

التوبيكس الي جات للناس في امتحان التـكـ:

- problem solving
- oop
- database

نبدأ بقا بالمطلوب

مطلوب تقرا المقالات الي هنا هي بتتكلم عن الكومبايلر وملخص
لقصة البرمجه بالاضافه لمقدمه بسيطه وده ترشيح منهم

https://www.tutorialspoint.com/computer_programming/index.htm

مطلوب نذاكريزك لغة برمجة معينه وترشيدات البعض والي انا
حذاكره هي ++c وكده كده الكونسبت واحد الاختلاف في
السينتاكس ودي المصادر:

ده مصدرين مرشحهم iti

المصدر الاول:

<https://www.coursera.org/learn/intro-programming>

المصدر الثاني:

<https://www.udacity.com/course/c-plus-plus-nanodegree--nd213>

طبعا الدنيا كتير في الي هما مرشدينه وغير البعض مننا ممكن
يواجهه مشكله مع الانجليزي فرشحتك انا الكورس ده مع
المهندس عادل نسيم

<https://www.youtube.com/playlist?list=PLClnYL3l2AajFAiw4s1U4QbGszcQ-rAb3>

اتعلمت وظبطت حالك دي بقا اسالة mcq علي اساسيات اللغه الي
اتعلمناها عشان تتعود علي شكل الامتحان

https://www.sanfoundry.com/cplusplus-interview-questions-answers/#google_vignette

<https://cranesvarsity.com/cplusplus-programming-mcq-interview-questions/>

<https://www.cppbuzz.com/c++/mcqs/c++-multiple-choice-questions>

**مطلوب منا نذاكر oop وهما مش حاطين مصدر فانا حطلك احسن
مصدرين**

المصدر الاول شرح الدكتور محمد الدسوقي :

<https://www.youtube.com/playlist?list=PL1DUmTEdeA6KLEvIO0NyrkT91BVle8BOU>

المصدر الثاني تبع iti وهو كورس علي مهاره تك:

<https://maharatech.gov.eg/course/view.php?id=2206>

دي بقا سلايدز ممكن تذاكر منها النظري برضو تبع الدكتور محمد الدسوقي

https://psauedusa-my.sharepoint.com/personal/m_eldesouki_psau_edu_sa/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Fm%5Feldesouki%5Fpsau%5Fedu%5Fsa%2FDocuments%2FJavaBooks%2FC%2B%2B%2FObject%20Oriented%20Programming%20in%20c%2B%2B%2Epdf&parent=%2Fpersonal%2Fm%5Feldesouki%5Fpsau%5Fedu%5Fsa%2FDocuments%2FJavaBooks%2FC%2B%2B&ga=1

دي مخلصات سريعه كده لو عاوز زتونة oop كنظري سريعا

<https://drive.google.com/file/d/1DnB1Hi3QzEpZwKyKbSEI-TlaISLAM7Yt/view?usp=sharing>

https://drive.google.com/file/d/1epJYHNPBwdoA8X0yXjzRQjZkZTDA_6jI/view?usp=sharing

شوية اساله mcq علي oop عشان تتعود علي شكل الامتحان

<https://www.javatpoint.com/oops-mcq>

<https://www.examveda.com/computer-science/practice-mcq-question-on-object-oriented-programming-using-c-plus-plus/>

<https://www.indiabix.com/cpp-programming/oops-concepts/>

<https://www.placementpreparation.io/mcq/object-oriented-programming/>
(مهم جد جدا)

مطلوب نذاكر داتا بيز

ترشيح iti مرشحين مصدرين:

اول مصدر من خان اكاديمي

<https://www.khanacademy.org/computing/computer-programming/sql>

المصدر الثاني كورس مهاره تك وده ارشدهولك جد جدا يعني

<https://maharatech.gov.eg/course/view.php?id=740>

مصادر بقا مني ومن الزملاء وترشيحهم

اولا حذاكر ده وهو لامم بعض النظري عن الداتا بيز

<https://youtu.be/rQKJFIsifR8?si=1RQl0fll8dmQwXoY>

حذاكر لحد الفيديو ال 12 هنا

https://www.youtube.com/playlist?list=PLe_UJpVeP8qCHeTPVPc2gQXuQr7

[AniUbn](#)

بعدين حذاكر كورس مهاره تك انا اصلا شايفه كافي لكل حاجه حتي يغني

عن الي فات

<https://maharatech.gov.eg/course/view.php?id=740>

دي تطبيقات عليها

<https://www.youtube.com/playlist?list=PLxbVBWjVdAEj8TmOUKPG0avUmlqSoQOpf>

دي ملخصات سريعة للنظري

https://drive.google.com/file/d/1E0OvRg_SsBrfc3wxy3Wxes9IRUAZgZ6L/view?usp=sharing

<https://drive.google.com/file/d/1GNBCqbh-fhyP3iFO7NJKf2fcVE1ZPaJv/view?usp=sharing>

ملخص لكل الكويري وجزء الكود

https://drive.google.com/file/d/1hKLnIMM18WAB0Uyb0ng1vzk_U7yOFmuls/view?usp=sharing

دي بقا اسالة mcq للتدريب علي شكل الامتحان

<https://www.includehelp.com/sql/mcq.aspx>

<https://www.interviewbit.com/dbms-mcq/>

<https://www.javatpoint.com/sql-mcq>

<https://www.sanfoundry.com/sqlserver-mcqs-basic-sql-1/>

<https://www.interviewbit.com/sql-mcq/>

https://www.sanfoundry.com/database-mcqs-sql-queries/#google_vignette

<https://www.placementpreparation.io/mcq/sql/>

https://www.w3schools.com/sql/sql_quiz.asp

مطلوب منا نذاكر html css

ده مصدر ترشيح من iti من مهاره تك

https://maharatech.gov.eg/enrol/index.php?id=36&utm_source=facebook&utm_medium=social&utm_campaign=launch&utm_content=image&utm_term=facebook-html

ده بقا ترشيحي ذاكر من هنا html css

<https://youtube.com/playlist?list=PLU0wE7dsJI8SPVnK6zi3XhxyYKfVFK7Oc&si=7FHcTHhul-Oa0hMs>

ملخص html مع الشرح

<https://drive.google.com/file/d/1wSgAE1Wnd-tUbw3uzptlIMUiPJzxs2qa/view?usp=sharing>

ملخص css مع الشرح

https://drive.google.com/file/d/1n_h10Ygymc9IRIG06sO-HEy6hc998KRY/view?usp=sharing

نيجي لجزء اسالة mcq عشان تدرب علي شكل الامتحان

html

<https://www.javatpoint.com/html-mcq>

<https://www.w3schools.com/quiztest/quiztest.asp?qtest=HTML>

<https://www.examveda.com/html/practice-mcq-question-on-basic-html/>

css

<https://www.w3schools.com/quiztest/quiztest.asp?qtest=CSS>

<https://www.javatpoint.com/css-mcq>

https://www.examveda.com/css/practice-mcq-question-on-basic-css/#google_vignette

لو عندك وقت وهنا اختلف الاراء علي جزء الجافا اسكريبت بس انا شايف هي لغة برمجه وانت من الي فوق بقا عندك الكونسبت فاقرا عن الفروق لانه مفيش وقت لكورس وارشحك تذاكر من هنا

https://drive.google.com/drive/folders/1nweZ9gZBRqqgPfdPCLJbJPrfuAARV_8e?usp=drive_link

وده مصدر ثاني برضو

<https://www.w3schools.com/js/>

وده شوية اساله mcq علي جافا اسكريبت

<https://www.javatpoint.com/javascript-mcq>

<https://www.sanfoundry.com/1000-javascript-questions-answers/>

تجارب الزملاء الدفعه الي فانت

https://docs.google.com/document/d/1rkx_1mPDLsGoZCZl4SlI3DxcfmRKS_NfG/edit?usp=sharing&ouid=109670532343611538825&rtpof=true&sd=true

اساله اتكررت في الانترفيو الدفعه الي فانت

https://drive.google.com/drive/folders/1lLdYSZp00JZKmT66iNv_kPSMxUmS8i8?fbclid=IwAR3E5ba4W4SDtZgsoGT4S-aVNRkZbdltciEqg86zOxiX_envH_UWr-hRGOJQ

اساله اتكررت في امتحان السنة دي

1. What is the primary use of a loop in programming?

- A) To reduce the length of the code.
- B) To execute a block of code once.
- C) To execute a block of code multiple times.
- D) To prevent errors in the code.

Answer: C) To execute a block of code multiple times.

Explanation: Loops are primarily used to repeat a block of code multiple times, depending on a condition or the number of iterations required.

2. Which of the following is a disadvantage of using loops?

- A) They make the code more efficient.
- B) They can lead to infinite loops.
- C) They reduce the complexity of the code.
- D) They are difficult to understand.

Answer: B) They can lead to infinite loops.

Explanation: One of the disadvantages of loops is the potential for creating infinite loops if the termination condition is not correctly defined.

3. How do loops contribute to the scalability of code?

- A) By making the code more complex.
- B) By reducing the amount of code.
- C) By allowing the code to handle larger datasets without increasing code size.
- D) By making the code run faster.

Answer: C) By allowing the code to handle larger datasets without increasing code size.

Explanation: Loops help in scaling the code as they can process larger datasets without the need for additional code, making the program more efficient.

4. Which scenario is NOT ideal for using a loop?

- A) Iterating through a list of items.
- B) Repeating a task until a condition is met.
- C) Executing a block of code a fixed number of times.
- D) Performing a single operation on a variable.

Answer: D) Performing a single operation on a variable.

Explanation: Loops are generally used for repetitive tasks. For a single operation, a loop is unnecessary and can add unnecessary complexity.

5. What is a common issue that can occur with nested loops?

- A) They simplify the code.
- B) They always improve performance.
- C) They can lead to performance bottlenecks.
- D) They reduce code readability.

Answer: C) They can lead to performance bottlenecks.

Explanation: Nested loops, especially when dealing with large datasets, can significantly slow down the program, leading to performance issues.

6. What happens if you forget to update the loop variable in a while loop?

- A) The loop will execute exactly once.
- B) The loop will never start.
- C) The loop will result in a syntax error.
- D) The loop may run infinitely.

Answer: D) The loop may run infinitely.

Explanation: If the loop variable is not updated, the loop condition may never become false, leading to an infinite loop.

7. Which of the following is true about a `do-while` loop?

- A) It executes at least once before checking the condition.
- B) It never checks the condition.
- C) It may never execute the loop body.
- D) It is the same as a `while` loop.

Answer: A) It executes at least once before checking the condition.

Explanation: A `do-while` loop always executes the loop body at least once before evaluating the condition at the end of the loop.

8. What is the purpose of a `break` statement inside a loop?

- A) To exit the loop immediately.
- B) To pause the loop and resume later.
- C) To skip the current iteration and move to the next one.
- D) To restart the loop from the beginning.

Answer: A) To exit the loop immediately.

Explanation: The `break` statement is used to immediately terminate the loop and exit from it, regardless of the loop condition.

9. In which of the following scenarios would a loop be most useful?

- A) Checking if a number is positive or negative.
- B) Sorting a list of numbers.
- C) Repeatedly prompting a user for input until valid data is entered.
- D) Calculating the square root of a number.

Answer: C) Repeatedly prompting a user for input until valid data is entered.

Explanation: A loop is ideal for situations where a task needs to be repeated until a specific condition is met, such as validating user input.

10. Which loop type is best suited for iterating over a collection of items, such as elements in an array?

- A) while loop
- B) for loop
- C) do-while loop
- D) if statement

Answer: B) for loop.

Explanation: A **for** loop is particularly well-suited for iterating over collections, especially when the number of elements is known, as it allows easy access to each element by index.

11. What is a potential risk of using nested loops?

- A) Simplified code structure.
- B) Reduced readability of the code.
- C) Decreased likelihood of syntax errors.
- D) Increased speed of execution.

Answer: B) Reduced readability of the code.

Explanation: Nested loops can make the code harder to read and understand, especially if there are multiple levels of nesting.

12. How can you avoid an infinite loop in a while loop?

- A) By not using the loop at all.
- B) By ensuring that the loop condition eventually becomes false.
- C) By using a **break** statement after the first iteration.
- D) By setting the loop condition to always be true.

Answer: B) By ensuring that the loop condition eventually becomes false.

Explanation: To avoid an infinite loop, it's crucial to ensure that the loop condition will eventually become false, allowing the loop to terminate.

13. Which key uniquely identifies a record in a database table?

- A) Foreign Key
- B) Alternate Key
- C) Primary Key
- D) Composite Key

Answer: C) Primary Key

Explanation: The primary key is used to uniquely identify each record in a table.

14. What is the role of a foreign key in a database?

- A) To uniquely identify each record.
- B) To link two tables together.
- C) To create a composite key.
- D) To ensure high cardinality.

Answer: B) To link two tables together.

Explanation: A foreign key is used to establish a relationship between two tables, ensuring referential integrity.

15. What does high cardinality mean in the context of a database column?

- A) The column has many duplicate values.
- B) The column has many unique values.
- C) The column is used as a foreign key.
- D) The column can have null values.

Answer: B) The column has many unique values.

Explanation: High cardinality refers to a column with a large number of unique values, such as an ID number or email address.

16. Which type of participation ensures that every instance of an entity is involved in a relationship?

- A) Partial Participation
- B) Total Participation
- C) High Cardinality
- D) Composite Key

Answer: B) Total Participation

Explanation: Total participation means that all instances of the entity must be part of the relationship.

17. What is a composite key?

- A) A key that uniquely identifies a record with a single attribute.
- B) A combination of two or more attributes that uniquely identify a record.
- C) A foreign key in another table.
- D) A key with only unique values.

Answer: B) A combination of two or more attributes that uniquely identify a record.

Explanation: A composite key is used when no single attribute can uniquely identify a record, so a combination of attributes is used instead.

18. In database relationships, what does a one-to-many cardinality indicate?

- A) One record in one table is associated with exactly one record in another table.
- B) One record in one table can be associated with multiple records in another table.
- C) Multiple records in one table are associated with one record in another table.
- D) Multiple records in one table are associated with multiple records in another table.

Answer: B) One record in one table can be associated with multiple records in another table.

Explanation: A one-to-many relationship means that a single record in one table can relate to multiple records in another table.

19. What is an alternate key?

- A) A key that is always chosen as the primary key.
- B) A candidate key that was not chosen as the primary key.
- C) A key that ensures referential integrity.
- D) A key that consists of multiple attributes.

Answer: B) A candidate key that was not chosen as the primary key.

Explanation: An alternate key is a candidate key that could have been chosen as the primary key but was not selected.

20. What type of participation is indicated when not all instances of an entity need to be involved in a relationship?

- A) Total Participation
- B) Partial Participation
- C) High Cardinality
- D) Unique Key

Answer: B) Partial Participation

Explanation: Partial participation means that only some instances of the entity are required to participate in the relationship.

21.What is the correct execution order of SQL clauses in a query?

- A) `SELECT → WHERE → GROUP BY → ORDER BY → HAVING`
- B) `FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY`
- C) `SELECT → FROM → WHERE → GROUP BY → HAVING → ORDER BY`
- D) `SELECT → FROM → HAVING → GROUP BY → WHERE → ORDER BY`

Answer: B

22.Given the following query, what will be the result if the

`DISTINCT` keyword is omitted?

```
SELECT DISTINCT department, COUNT(employee_id)
FROM employees
GROUP BY department;
```

- A) The query will return all rows, including duplicates.

- B) The query will return the same result as with **DISTINCT**.
- C) The query will return an error.
- D) The query will return only unique departments without counting employees.

Answer: B

23. What does the **JOIN clause do in SQL?**

- A) Combines rows from two or more tables based on a related column.
- B) Removes rows from one table based on another table.
- C) Orders the result set based on one or more columns.
- D) Filters rows in the result set based on a condition.

Answer: A

24. What will be the output of the following SQL query?

```
SELECT name, salary  
FROM employees  
WHERE salary BETWEEN 3000 AND 5000  
ORDER BY salary DESC;
```

- A) Employees with salaries between 3000 and 5000, ordered by salary ascending.

- B) Employees with salaries between 3000 and 5000, ordered by salary descending.
- C) All employees ordered by salary descending.
- D) All employees with salaries between 3000 and 5000, ordered by name.

Answer: B

25. In which scenario would you use the **HAVING clause?**

- A) To filter rows before grouping them.
- B) To filter groups after aggregating them.
- C) To specify columns to be selected.
- D) To join multiple tables.

Answer: B

26. What is the output of the following code?

```
#include <stdio.h>

int main() {
    char p1[] = "greek";
    char *p2 = p1;

    printf("%s\n", p2);
    return 0;
}
```

- A) greek
- B) p1
- C) Compilation error
- D) greekp1

Correct Answer: A) greek

27. What will happen if you use the following `printf` statements with incorrect data types?

```
printf("%d", ay haga); // Suppose 'ay haga' is a string  
printf("%s", ay haga); // Suppose 'ay haga' is an integer
```

- A) Both lines will print the correct values.
- B) The first line will cause undefined behavior because `%d` expects an integer, but a string is passed.
- C) The second line will print the integer value correctly using `%s`.
- D) Both lines will result in a compilation error.

Correct Answer: B) The first line will cause undefined behavior because `%d` expects an integer, but a string is passed.

28. What will be the output of this code?

```
int array[3] = {1, 2};  
printf("array[2] = %d\n", array[2]);
```

- A) 1
- B) 2
- C) 0
- D) Compilation error

Correct Answer: C) 0

Explanation: In C++, when you create an array with a specified size and initialize only some of its elements, the uninitialized elements are automatically set to 0. Therefore, array[2] (the third element in the array) will be 0, so the output will be 0.

29. What is the output of the following C code?

```
#include <stdio.h>  
  
int main() {  
    printf("bank");  
    return 0;  
    printf("currency");  
}
```


- A) bank
- B) currency
- C) bankcurrency
- D) Nothing is printed

Correct Answer: A) bank

Explanation: The `printf("bank");` statement prints "bank" to the output. After `return 0;`, the program terminates immediately, so any code after `return` will not be executed.

30. Where are objects stored in a computer?

- A) RAM
- B) ROM
- C) Cache
- D) HDD

Correct Answer: A) RAM

Explanation: Objects are typically stored in RAM (Random Access Memory) during program execution. ROM (Read-Only Memory) and HDD (Hard Disk Drive) are used for different types of storage, and Cache is a smaller, faster memory used to speed up access to frequently used data.

31. What does "mathematical codes" refer to?

- A) Codes used to perform mathematical operations and algorithms in programming.
- B) Special symbols used in mathematics to represent complex equations.
- C) Encoded messages that use mathematical formulas for encryption.
- D) Color codes used in mathematical graphs and charts.

Correct Answer: A) Codes used to perform mathematical operations and algorithms in programming.

Explanation: "Mathematical codes" typically refer to programming codes that are designed to perform mathematical operations and algorithms. They help in solving mathematical problems and implementing mathematical concepts in software.

32. What is the difference between a primary key and a candidate key?

- A) A primary key is a unique identifier for a record in a table, while a candidate key is any column or set of columns that can uniquely identify records.
- B) A primary key is a column that can contain null values, while a candidate key cannot.
- C) A primary key can be used to join tables, while a candidate key cannot.
- D) A primary key is always selected from the candidate keys, while a candidate key is a potential primary key that is not necessarily selected.

Correct Answer: D) A primary key is always selected from the candidate keys, while a candidate key is a potential primary key that is not necessarily selected.

Explanation: A candidate key is any column or set of columns that can uniquely identify records in a table. The primary key is one of the candidate keys chosen to uniquely identify records and enforce entity integrity. The other candidate keys remain as unique identifiers but are not used as the primary key.

33. What does "the size of the class" refer to in programming?

- A) The number of data members and methods declared within the class.
- B) The amount of memory allocated for an instance of the class.
- C) The number of objects created from the class.
- D) The number of lines of code in the class definition.

Correct Answer: B) The amount of memory allocated for an instance of the class.

Explanation: The "size of the class" typically refers to the amount of memory that is allocated for an instance of the class. This size includes the memory required for all its data members and any overhead needed by the class.

34. Does a class have a size in programming?

- A) Yes, the size of a class is determined by the amount of memory allocated for an instance of the class.
- B) No, a class does not have a size; it is just a blueprint for creating objects.
- C) Yes, the size of a class is determined by the number of methods it contains.
- D) No, the size of a class depends on the number of objects created from it.

A) Yes, the size of a class is determined by the amount of memory allocated for an instance of the class.

Explanation:

In programming, a class itself doesn't occupy memory until an object (or instance) is created. The size of the class is essentially the size of the memory allocated for an object of that class, which depends on the size of its member variables (both data members and inherited members, if applicable). Methods do not directly affect the size of a class instance, but virtual methods may add some overhead for a virtual table (vtable) if polymorphism is involved.

35. What is the size of a class in programming?

- A. Size of its data members
- B. Size of its data members and parent class data members
- C. Has no size

Correct Answer: Size of its data members and parent class data members

Explanation: The size of a class is determined by the memory allocated for its data members as well as the data members inherited from its parent class. This total size reflects the amount of memory required for an instance of the class.

36. What is the output of the following C++ code?

```
#include <cstdio>

void Show() {
    printf("vamos");
}

int main() {
    Show();
    printf("Argentina");
    return 0;
}
```

- A) vamosArgentina
- B) Argentinevamos
- C) vamos
- D) Argentina

Answer: A) vamosArgentina

Explanation: The Show function is called first in the main function, which prints vamos.

After that, printf("Argentina"); is executed, printing Argentina.

The final output combines both printed strings, resulting in vamosArgentina.

37. Which of the following is not an access modifier in Java?

- A) public
- B) private
- C) protected
- D) internal

Answer and: D) internal

Explanation: In Java, internal is not an access modifier. The valid access modifiers in Java are public, private, protected, and the default access level (no modifier). The internal keyword is used in other programming languages, like C#, but not in Java.

38. What is the access modifier not in Java?

- A. Public
- B. Private
- C. Protected
- D. Package

correct answer :Package.

Explanation:

In Java, the access modifiers are:

Public: Accessible from any other class.

Private: Accessible only within the class it is declared.

Protected: Accessible within the same package and by subclasses.

Package is not an official access modifier; however, if no access modifier is specified, the default access level (also known as package-private) is applied, meaning it is accessible only within the same package.

مهم الجزء ده مهم

Overriding

- Method Overriding: تغيير تنفيذ دالة في الفئة الفرعية.
- Virtual Function: الدالة التي يمكن تجاوزها في الفئة الأساسية.
- Base Class Method: الدالة التي يتم تجاوزها في الفئة الأساسية.
- Derived Class Method: الدالة التي تعيد تعريفها في الفئة الفرعية.
- Dynamic Polymorphism: التعددية الشكلية التي تتحقق عبر التجاوز، حيث يتم تحديد الدالة التي سيتم استدعاؤها عند التنفيذ.

Overloading

- Method Overloading: تعريف عدة دوال بنفس الاسم ولكن مع توقيعات مختلفة.
- Function Overloading: نفس معنى Overloading، ولكن يُستخدم للإشارة إلى الدوال.
- Compile-time Polymorphism: التعددية الشكلية التي تتحقق عبر التحميل، حيث يتم تحديد الدالة التي سيتم استدعاؤها عند الترجمة بناءً على التوقيع.

39.

Q33 Which of the following best describes "dynamic polymorphism"?

☐ A Compile-time method binding

☐ B The ability to create dynamic classes at runtime

☐ C Method overriding where method calls are resolved at runtime

☐ D The creation of objects at runtime

[View Answer](#)

[Report](#)

Correct Answer: C) Method overriding where method calls are resolved at runtime

Explanation:

- Dynamic Polymorphism is a type of polymorphism in object-oriented programming where the method to be invoked is determined at runtime. This is achieved through method overriding, where a subclass provides a specific implementation of a method that is already defined in its superclass. The exact method that gets called is resolved when the program runs, hence it's "dynamic."
- Compile-time method binding (option A) refers to static polymorphism (e.g., method overloading) which is resolved at compile time.
- Option B refers to dynamic class creation, which is unrelated to polymorphism.
- Option D relates to object creation, not directly to dynamic polymorphism.

40-

9. Under which pillar of OOPS do base class and derived class relationships come?

1. Polymorphism
2. Inheritance
3. Encapsulation
4. Abstraction

41-

25. What happens when an object is passed by reference?

- a) Destructor is called at end of function
- b) Destructor is called when called explicitly
- c) Destructor is not called
- d) Destructor is called when function is out of scope

^ View Answer

Answer: c

42-

48. Which among the following is not a necessary condition for constructors?

- a) Its name must be same as that of class
- b) It must not have any return type
- c) It must contain a definition body
- d) It can contains arguments

^ View Answer

Answer: c

43-

[View Answer](#)

13. What will be the output of the following JavaScript code?

```
var a = 0;
var b = 0;
while (a < 3)
{
    a++;
    b += a;
    console.log(b);
}
```

- a) 136
- b) 123
- c) 013
- d) 01

[^ View Answer](#)

Answer: a

Explanation: A while loop checks for the condition first before executing the loop statements. In this case, the variable a is incremented by 1 in every iteration till the loop breaks i.e., once the variable a becomes 3.

44-

15. What will be the output of the following JavaScript code?

```
let a = 0;  
for (a; a < 5; a++);  
console.log(a);
```

- a) 0
- b) error
- c) 4
- d) 5

 [View Answer](#)

Answer: d

Explanation: The output will be 5 because a increments from 0 to 5 within the loop before the loop condition fails. Remember, there is a semicolon immediately after the for loop, hence the variable a will simply increment up to 5.

45-

5. Which feature of OOP indicates code reusability?

- a) Abstraction
- b) Polymorphism
- c) Encapsulation
- d) Inheritance

 [View Answer](#)

Answer: d

46-What is the type of null in most programming languages?

- A) Object
- B) Undefined
- C) Null
- D) Integer

Correct Answer: C) Null

Explanation:

In most programming languages, null is a special value that represents the absence of any object or value. It is not an object itself but a placeholder for "no value" or "nothing." Some languages like JavaScript may show confusing behavior, where `typeof null` returns "object", but this is a historical quirk, and technically, null is not an object.

47- In JavaScript, what is the result of typeof null?

- A) Object
- B) Undefined
- C) Null
- D) Boolean

Correct Answer: A) Object

Explanation:

In JavaScript, typeof null returns "object". This is a historical quirk in the language. Although null represents the absence of a value or an empty reference, the typeof operator mistakenly identifies it as an object due to how early versions of JavaScript were implemented.

48- What does the async keyword do in JavaScript?

- A) It pauses the execution of a function until an event occurs.
- B) It allows a function to run asynchronously and always returns a promise.
- C) It creates a new thread for concurrent execution.
- D) It converts a synchronous function into an asynchronous one.

Correct Answer: B) It allows a function to run asynchronously and always returns a promise.

Explanation:

The `async` keyword in JavaScript is used to define asynchronous functions.

When a function is marked as `async`, it always returns a promise, even if the function doesn't explicitly return one. The `await` keyword can then be used inside an `async` function to pause its execution until a promise is resolved.

49- What will be the output of the following C code snippet?

```
int arr[][2][3] = {{1, 2, 3}, {4, 5, 5}};  
printf("%d, %d", arr[0][0][0], arr[1][1][0]);
```

- A) 1, 5
- B) 1, 4
- C) 4, 5
- D) Compilation error

Correct Answer: A) 1, 5

Explanation:

`arr[0][0][0]` refers to the first element of the first 2D array, which is 1.

`arr[1][1][0]` refers to the first element of the second row of the second 2D array, which is 5. Thus, the output is 1, 5.

50-

What will be the output of the following C code snippet?

c

 Copy code

```
int arr[][3] = {5, 6, 7};  
printf("%d", arr[0][1]);
```

- A) 5
- B) 6
- C) 7
- D) Compilation error

Correct Answer:

B) 6

Explanation:

- `arr[0][1]` refers to the element in the first row and second column of the array `arr`. In this case, the value stored in `arr[0][1]` is `6`. Therefore, the output will be `6`.

51-

What will be the output of the following C code snippet?

c

 Copy code

```
int arr[2][2][3] = {{{1, 2, 3}, {4, 5, 6}}, {{7, 8, 9}, {10, 11, 12}}};  
printf("%d, %d\n", arr[0][0][0], arr[1][1][1]);
```

- A) 1, 11
- B) 1, 10
- C) 4, 11
- D) 7, 12

Correct Answer:

A) 1, 11

Explanation:

- `arr[0][0][0]` refers to the first element in the first 3D sub-array, which is `1`.
- `arr[1][1][1]` refers to the second element in the second row of the second 3D sub-array, which is `11`. Thus, the output is `1, 11`.

في ناس دمغها ساحات وعاوزه شرح سريع للي حاصل في array

شرح الكود:

- `arr[0][0][0]` : يشير إلى الصف الأول.
- `arr[0]` : يشير إلى الصف الفرعي الأول في الصف الأول.
- `arr[0][0]` : يشير إلى العنصر الأول في الصف الفرعي الأول، وهو 1.
- `arr[1][1][1]` : يشير إلى الصف الثاني.
- `arr[1]` : يشير إلى الصف الفرعي الثاني في الصف الثاني.
- `arr[1][1]` : يشير إلى العنصر الثاني في الصف الفرعي الثاني، وهو 11.

52-

What will be the output of the following C code snippet?

c

Copy code

```
char arr[] = {'a', 'b', 'c', 'd'};
printf("%c", arr);
printf("%s", arr);
```

- A) Compilation error
- B) aabcd
- C) aabcd
- D) a

Correct Answer:

A) Compilation error

Explanation:

- `printf("%c", arr);` : This line attempts to print `arr` as a single character, but `arr` is an array of characters, not a single character. The `%c` format specifier expects a single `char` argument, not an array, leading to a compilation error.
- `printf("%s", arr);` : This line will print the contents of `arr` as a string. However, since `arr` lacks a null terminator (`'\0'`), this leads to undefined behavior and might print garbage values or cause a runtime error.



53-

What will be the output of the following C code snippet?

c

Copy code

```
char arr[] = {'a', 'b', 'c', 'd', '\0'};
printf("%c\n", arr[0]);
printf("%s\n", arr);
```

A)

a

abcd

B)

a

abcd

C)

a

a

D)

a

abc



Correct Answer:

A) a

abcd

Explanation:

`printf("%c\n", arr[0]);`: This line prints the first character in the array `arr`, which is `'a'`.

`printf("%s\n", arr);`: This line prints the entire array as a string. Since `arr` ends with a null terminator (`'\0'`), it will correctly print the characters `'a'`, `'b'`, `'c'`, `'d'` as a string.

54-

What is the output of this code snippet?

```
let arr = [10, 20, 30, 40];  
let newArr = arr.map(x => x / 10);  
console.log(newArr);
```

☒ A [1, 2, 3, 4] ☐ B [10, 20, 30, 40]

☐ C [0.1, 0.2, 0.3, 0.4] ☐ D Error

View Answer

55-

Which data type is used to represent all types of numbers, including integers and floating-point numbers, in most programming languages?

A) `int`
B) `float`
C) `double`
D) `number`

Correct Answer:
D) `number`

Explanation:

- `int` : Represents integer values (whole numbers) only.
- `float` : Represents floating-point numbers (decimal numbers) but may not have the precision needed for all applications.
- `double` : Represents floating-point numbers with double precision (greater accuracy) but still not suitable for all numeric requirements.
- `number` : This data type is used in languages like JavaScript to represent both integers and floating-point numbers with a single type. In most programming languages, you may need to use different data types or a combination of them to cover all numeric needs.

↓

56-

What is the access specifier that is commonly used for class entities to restrict access to their members and enforce encapsulation in object-oriented programming?

- A) `public`
- B) `private`
- C) `protected`
- D) `internal`

Correct Answer:

B) `private`

Explanation:

- `public` : Allows access to the class members from any part of the program.
- `private` : Restricts access to the class members to only within the class itself, enforcing encapsulation and hiding the implementation details from other classes.
- `protected` : Allows access to the class members from within the class and its derived classes (subclasses).
- `internal` : (Used in languages like C#) Restricts access to the class members to the current assembly (i.e., code within the same project or module).

The `private` access specifier is commonly used to protect the internal state of a class and restrict direct access from outside the class.



57-

In JavaScript, how are values passed to functions for primitive types (like numbers and strings) and non-primitive types (like objects and arrays)?

- A) All values are passed by value.
- B) All values are passed by reference.
- C) Primitive types are passed by value, while non-primitive types are passed by reference.
- D) Primitive types are passed by reference, while non-primitive types are passed by value.

Correct Answer:

C) Primitive types are passed by value, while non-primitive types are passed by reference.

Explanation:

- **Primitive types** (such as numbers, strings, booleans, null, and undefined) are passed by value, which means that a copy of the value is passed to the function. Changes to the parameter inside the function do not affect the original value.
- **Non-primitive types** (such as objects, arrays, and functions) are passed by reference, which means that a reference to the object is passed to the function. Changes to the object or array inside the function affect the original object or array.

This distinction is important for understanding how data is managed and manipulated within functions in JavaScript.

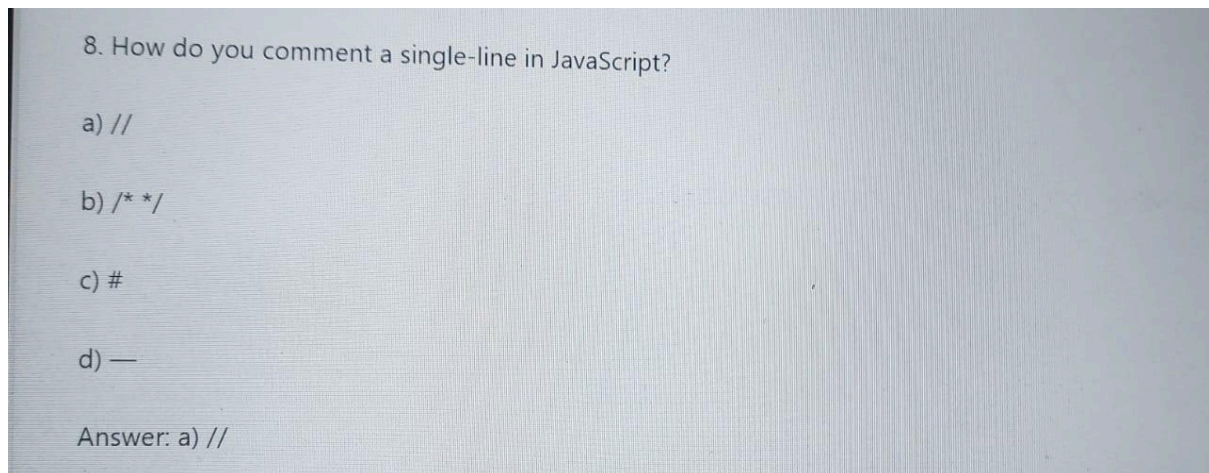
58-

4. What does the "===" operator in JavaScript do?

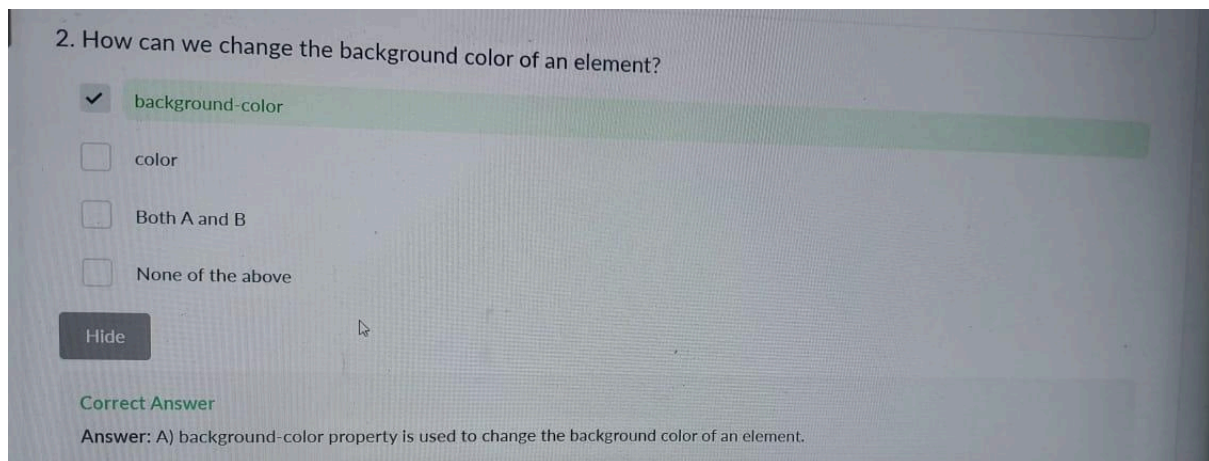
- a) Compares values for equality without type coercion
- b) Assigns a value to a variable
- c) Checks if a variable is defined
- d) Performs bitwise XOR operation

Answer: a) Compares values for equality without type coercion

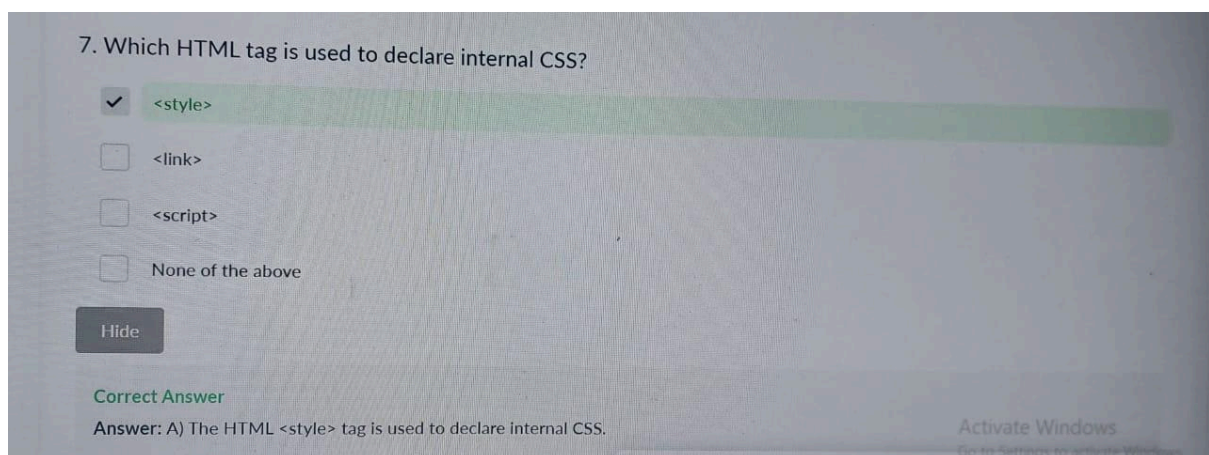
59-



60-



61-



62-

10. How can we write comments in CSS?

☒ `/* */`

☐ `//`

☐ `#`

☐ All of the above

Hide

Correct Answer

Answer: A) `/* */` is used to write comments in CSS.

63-

12. How do we write comments in HTML?

a) `</.....>`

b) `<!--.....-->`

c) `</...../..>`

d) `<.....!>`

^ View Answer

Answer: b

Explanation: Browser ignores comment in a code. There are always two types of command i.e. single line command and multiple line command. If one wants to add a comment in code, add the text between these characters `<!--.....comment.....-->`. It will not visible in the user's browser.

64-

1. The full form of CSS is:

☒ Cascading Style Sheets

☐ Coloured Special Sheets

☐ Color and Style Sheets

☐ None of the above

Hide

Correct Answer

Answer: A) CSS stands for Cascading Style Sheets.

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